

Certified Reductions of the Weight-6 Euler Sum at Dyadic Arguments: A Corrected Exact Closed Form at $x = 1/2$, a Certified Relation at $x = 1/4$, and the Boundary of the Dyadic Field

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Abstract

We study the weight-6 Euler sum

$$S_{4,2}(x) = \sum_{n \geq 1} \frac{H_{n-1}x^n}{n^5},$$

at dyadic arguments. This corrected revision withdraws the first-version claim of exact closed forms at all three arguments in a 21-constant basis. Those relations plateau near finite residuals and are not certified exact. The corrected result is: an exact closed form for $S_{4,2}(1/2)$ in a 13-element dyadic basis; a certified exact relation for $S_{4,2}(1/4)$ in a depth-2 multiple-polylogarithm basis; and an open status for $S_{4,2}(-1/2)$.

1 Correction note

The first version used a contaminated 21-element basis containing Clausen constants at $\pi/3$ and omitted the single irreducible depth-2 generator of the weight-6 level-2 space. A small residual at one precision is not evidence of exactness. This revision accepts a relation only when the residual tracks increasing precision and the rational coefficients remain stable.

2 Definitions

We use

$$S_{4,2}(x) = \sum_{n \geq 1} \frac{H_{n-1}x^n}{n^5} = \text{Li}_{5,1}(x, 1).$$

For $|x| \leq 1/2$, the depth-2 multiple polylogarithms used in this repository are evaluated by direct geometrically convergent summation.

3 Corrected exact closed form at $x = 1/2$

The implemented 13-element dyadic basis is

$$\zeta(6), \zeta(3)^2, \zeta(5)\log 2, \zeta(3)\log^3 2, \pi^2\zeta(3)\log 2, \pi^4\log^2 2, \pi^2\log^4 2, \log^6 2, \\ \text{Li}_6(1/2), \text{Li}_5(1/2)\log 2, \text{Li}_4(1/2)\log^2 2, \pi^2\text{Li}_4(1/2), S_{4,2}(-1).$$

The coefficient vector in this order is

$$-\frac{51}{32}, -\frac{1}{4}, -\frac{1}{32}, -\frac{1}{6}, \frac{1}{12}, \frac{1}{1440}, \frac{1}{144}, -\frac{1}{240}, 2, 1, 0, 0, -\frac{1}{2}.$$

4 Certified relation at $x = 1/4$

The implemented relation is

$$S_{4,2}(1/4) = \frac{49}{48}\zeta(3)^2 + \frac{7}{18}\zeta(3)\log^3 2 - \frac{7}{36}\pi^2\zeta(3)\log 2 + \frac{1}{108}\pi^4\log^2 2 \\ - \frac{1}{27}\pi^2\log^4 2 + \frac{1}{27}\log^6 2 + \frac{4}{9}\text{Li}_3(-1/2)\log^3 2 - \frac{2}{9}\pi^2\text{Li}_3(-1/2)\log 2 \\ + \frac{7}{3}\zeta(3)\text{Li}_3(-1/2) + \frac{8}{3}\text{Li}_{3,3}(-1/2) + 8\text{Li}_{4,2}(-1/2) + 16\text{Li}_{5,1}(-1/2) \\ - \frac{1}{6}\text{Li}_{3,3}(1/4) - \frac{1}{2}\text{Li}_{4,2}(1/4).$$

This is a certified exact relation, not a reduction to independently known constants, because it contains unreduced depth-2 MPLs including $\text{Li}_{5,1}(-1/2) = S_{4,2}(-1/2)$.

5 Open status at $x = -1/2$

No certified closed form for $S_{4,2}(-1/2)$ is implemented. The repository deliberately raises an error rather than evaluating a withdrawn relation.

6 Reproducibility

Run `pytest -q`. The tests compare the corrected dot-product evaluations against direct series evaluation and verify that the open case is not exposed as a closed form.